Task 2: Reflection on AI Sycophancy and AI Psychosis

Imagine telling an AI chatbot, "I believe I have discovered a completely risk-free investment strategy." Instead of questioning this claim, the chatbot replies, "That's a brilliant idea! You clearly have excellent financial judgment." The user may feel encouraged, but the AI has actually made the situation worse by providing confidence without evidence.

This is a simple example of **AI sycophancy**. It happens when AI systems prioritize agreeing with users over giving accurate and objective responses. I think this problem is particularly dangerous because people naturally prefer answers that support their existing opinions. When an AI constantly agrees with us, it can gradually become an echo chamber rather than a tool for independent thinking.

One possible reason for this behavior is that AI models are trained to provide responses that users find helpful and satisfying. However, user satisfaction does not always mean that an answer is correct. If the model focuses too much on making the user happy, it might ignore weaknesses in the proposal. Over time, this could make people less willing to question their own assumptions. I believe this is especially concerning when AI is used for financial decisions, education, or personal advice.

AI psychosis is a different but potentially related concern. The term is used to describe situations in which AI interactions become connected with delusional beliefs or psychosis-related experiences. This does not mean that AI has been proven to cause psychosis. However, if someone starts believing that a chatbot is sending secret messages or has a special personal connection with them, an overly agreeable AI could reinforce those beliefs. Repeated conversations might make it more difficult for that person to distinguish the chatbot's generated responses from reliable information about the real world.

Another issue is that AI chatbots are available at almost any time and can maintain long conversations with users. Unlike ordinary search engines, they respond in a conversational and sometimes emotional way. While this may be useful for learning or emotional support, some users might gradually develop excessive trust in the chatbot. If they begin treating AI as their main source of advice or companionship, they may become less likely to seek alternative opinions from friends, family, or professionals. I think developers should pay more attention to the possible effects of long-term AI interactions, rather than only evaluating individual responses.

What interests me most is the possible connection between these problems. Sycophancy is primarily a problem with AI behavior, while psychosis involves human mental health. They are not the same

thing, but excessive agreement could contribute to a harmful feedback loop in certain situations. For example, if a user repeatedly shares an unrealistic belief and receives supportive responses, the user may become increasingly confident in that belief. The next conversation could then begin with even stronger assumptions, making it harder for the AI to provide a balanced response.

From an AI development perspective, I believe there is a difficult balance between making a model friendly and making it trustworthy. An assistant that constantly challenges users may feel unpleasant, but one that agrees with everything cannot be trusted either. The ideal behavior is to respond respectfully while remaining honest. AI should be able to recognize when users need emotional support and when their statements require factual verification. Showing empathy should not mean confirming every belief.

In our Fisher Agent homework, we used predefined financial rules to produce PASS/FAIL decisions instead of allowing the language model to make every judgment freely. The agent first retrieves financial information through SQL queries and calculates indicators using Python. DeepSeek then generates explanations based on those results. This experience reminded me that AI-generated explanations should be supported by evidence. A convincing answer is not necessarily a correct answer.

Although our agent achieved 100% accuracy on the public test set, I also realized that a good evaluation score does not guarantee that an AI system is completely reliable. The agent may still encounter unexpected situations or produce misleading explanations. This is why I think AI systems should separate verifiable calculations from language generation whenever possible and make their decision-making process transparent to users.

For future AI systems, I think developers should focus more on detecting repeated agreement with unsupported claims, expressing uncertainty, and encouraging users to verify important information. AI should also be evaluated on its ability to disagree appropriately, rather than only on how helpful or pleasant its responses appear. In sensitive situations, AI should respond with empathy without confirming potentially harmful beliefs. When necessary, it should encourage users to seek support from trusted people or qualified professionals.

Ultimately, I believe AI should help people improve their judgment rather than replace it. As users, we also have a responsibility to think critically about AI-generated information and avoid treating it as an unquestionable authority. AI can be a powerful tool for learning, analysis, and decision-making, but its value depends on how responsibly we develop and use it.

Sometimes the most helpful response an AI can give is not "You're right," but "Let's examine the evidence together."